

MAINTENANCE OPERATIONS CLIENT

Data Validation & Dashboard Testing

Power BI Quality Assurance

Document Control	Detail
Engagement	Aug-Nov 2024
Data Scope	Approximately 10,000 historical records across 20 teams
Solution	Power BI dashboard
Testing Scope	Data quality, KPI logic, filters, reconciliation and dashboard behaviour
Status	Portfolio testing specification based on confirmed project rules

Evidence boundary: The engagement included testing/validation work, but exact historical test-case IDs, execution dates, defect counts and formal UAT sign-off are not retained. The cases below are a reproducible portfolio test specification.

1. Data Validation

Test ID	Validation	Expected Result
DV-01	Check duplicate maintenance records.	Duplicates are identified and controlled before KPI calculation.
DV-02	Check blank records.	Blank/non-usable records do not inflate analytical counts.
DV-03	Check team-name consistency.	Equivalent team names are standardized.
DV-04	Check status-name consistency.	Status values map consistently to governed reporting categories.
DV-05	Check date formats.	Created, Due and Completion dates are valid and consistently interpreted.
DV-06	Check missing values in analytical fields.	Missing values remain transparent; unsupported values are not fabricated.
DV-07	Check asset/equipment, category, priority and location labels.	Values are standardized where supported by source evidence.
DV-08	Check prepared record volume against source extracts.	Material record loss or unexpected duplication is investigated.

2. KPI & Business Rule Testing

Test ID	Test	Expected Result
KPI-01	Recalculate Completion Rate.	Completed Maintenance Tasks / Total Assigned Maintenance Tasks x 100.
KPI-02	Validate retained overall result.	Completion Rate reconciles to 92% for the retained engagement scope.
KPI-03	Validate overdue classification.	Every overdue record satisfies Due Date < Current Date AND Status != Completed.
KPI-04	Check completed records against overdue.	Completed records are excluded from overdue.
KPI-05	Check invalid/missing Due Date.	Records without a valid Due Date are not falsely classified as overdue.
KPI-06	Validate team completion.	Team views reuse the same governed Completion Rate logic.
KPI-07	Validate completion time.	Only records meeting the documented reconstructed date rule are included.
KPI-08	Validate backlog.	One documented reconstructed backlog definition is used consistently before implementation.
KPI-09	Validate high-priority pending.	Documented priority and status mappings are applied consistently.

92% control: The 92% value is a retained analytical result, not a hard-coded dashboard value or a claimed performance improvement.

3. Dashboard Functional Testing

Test ID	Scenario	Expected Result
FT-01	Change Team filter.	Relevant KPIs and visuals update to the selected team.
FT-02	Change Technician filter.	Relevant technician-attributed visuals update consistently.
FT-03	Change Priority filter.	Priority-dependent measures and visuals update.
FT-04	Change Category / Asset filters.	Relevant operational views update without affecting unrelated logic.
FT-05	Apply equivalent filters across pages.	Equivalent measures reconcile under the same filter context.
FT-06	Clear filters.	Dashboard returns to the full available analytical scope.
FT-07	Review missing/unknown values.	Unknown values remain visible or governed; they are not silently reassigned.

4. Dashboard Reconciliation

Area	Reconciliation Check
Completion Rate	Dashboard measure matches governed calculation.
Completed / Open / Pending	Visual totals reconcile with standardized status data.
Overdue	Dashboard count reconciles with record-level confirmed overdue logic.
Team / Technician	Aggregated views reconcile with underlying filtered maintenance records.
Priority	Priority visuals reconcile with standardized priority values.
Category / Asset	Breakdowns reconcile with prepared source fields.
Backlog	All visuals use the same reconstructed backlog definition.

5. Negative & Edge-Case Testing

Scenario	Expected Behaviour
Completion Date before Created Date	Record is flagged/excluded from reconstructed completion-time calculation.
Missing Due Date	Record is not automatically classified as overdue.
Completed status with past Due Date	Record is not overdue under the confirmed rule.
Inconsistent team/status label	Standardization occurs before reporting.
Missing technician attribution	Record is not assigned to a fabricated technician.
Duplicate work-order/task record	Duplicate control prevents inflated KPI results.
No records after filtering	Visuals return blank/zero behaviour without misleading values.

6. Acceptance Checklist

Check	Acceptance Condition
Data quality	Known duplicate, missing-value, naming and date-format issues are addressed transparently.
Completion Rate	Formula is consistent and retained overall result reconciles to 92%.
Overdue	Confirmed overdue rule is applied consistently.
Filters	Relevant visuals respond correctly.
Cross-page consistency	Equivalent measures reconcile under equivalent filters.
Backlog	Reconstructed definition is documented before use.
Confidentiality	Public portfolio outputs do not expose confidential client or personnel information.
Illustrative values	Synthetic dashboard values are not represented as historical client results.

7. Testing Outcome

The portfolio QA approach verifies the path from prepared maintenance data to KPI calculation and dashboard behaviour. It focuses on the rules most material to the case study: completion, overdue classification, status consistency, filter behaviour and reconciliation.

Historical boundary: Exact historical defect counts, pass/fail logs and formal UAT approval records are not retained and are therefore not claimed.